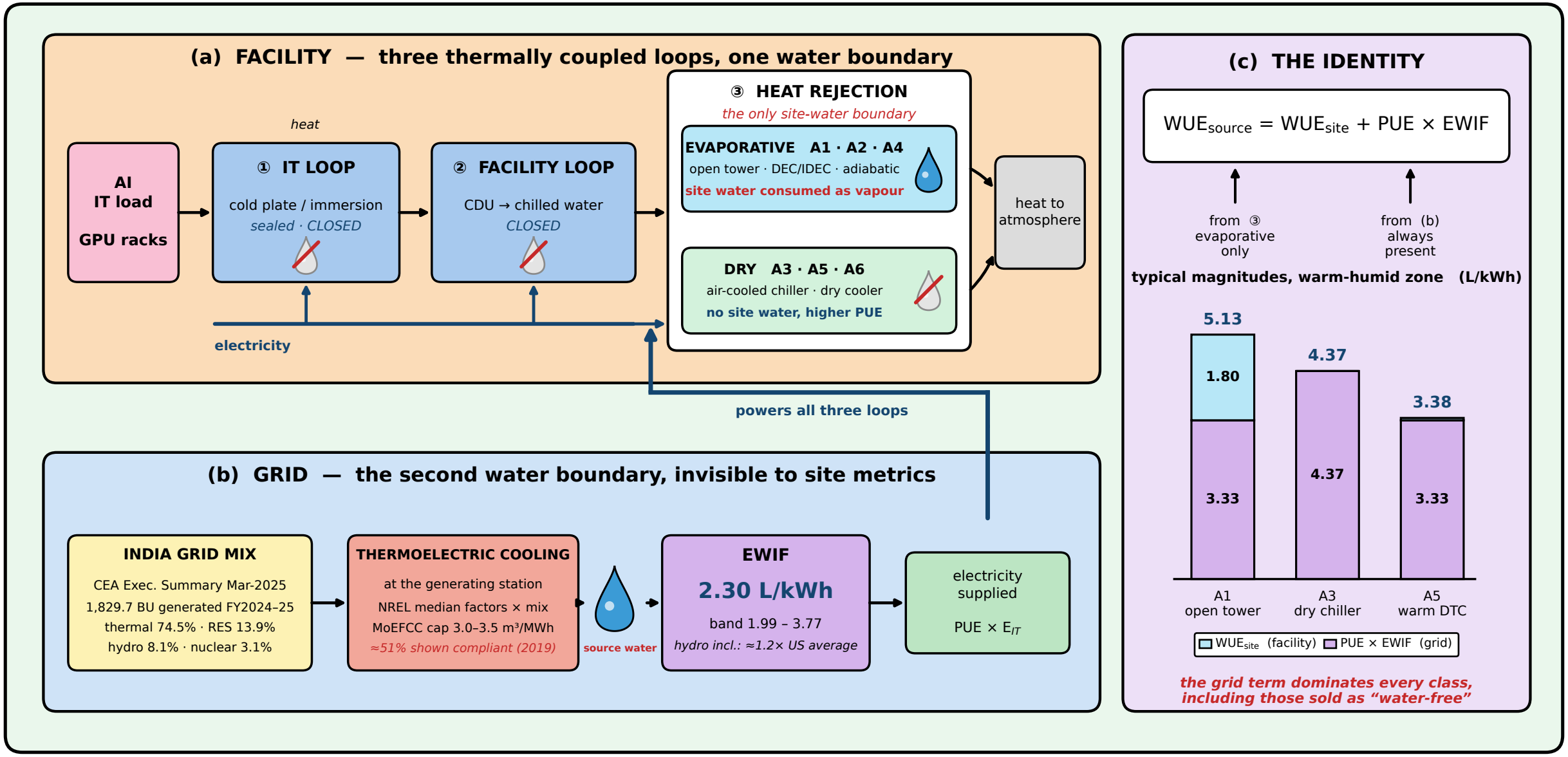

(a) FACILITY — three thermally coupled loops, one water boundary
AI
IT load
GPU racks
heat
① IT LOOP
cold plate / immersion
sealed · CLOSED
② FACILITY LOOP
CDU → chilled water
CLOSED
③ HEAT REJECTION
the only site-water boundary
EVAPORATIVE A1 · A2 · A4
open tower · DEC/IDEC · adiabatic
site water consumed as vapour
DRY A3 · A5 · A6
air-cooled chiller · dry cooler
no site water, higher PUE
heat to atmosphere
electricity
powers all three loops
(b) GRID — the second water boundary, invisible to site metrics
INDIA GRID MIX
CEA Exec. Summary Mar-2025
1,829.7 BU generated FY2024–25
thermal 74.5% · RES 13.9%
hydro 8.1% · nuclear 3.1%
THERMOELECTRIC COOLING
at the generating station
NREL median factors × mix
MoEFCC cap 3.0–3.5 m³/MWh
≈51% shown compliant (2019)
source water
EWIF
2.30 L/kWh
band 1.99 – 3.77
hydro incl.: ≈1.2× US average
electricity supplied
PUE × E_IT
(c) THE IDENTITY
WUE_source = WUE_site + PUE × EWIF
from ③ evaporative only
from (b) always present
typical magnitudes, warm-humid zone (L/kWh)
5.13
1.80
3.33
4.37
4.37
3.38
3.33
A1 open tower
A3 dry chiller
A5 warm DTC
WUE_site (facility)
PUE × EWIF (grid)
the grid term dominates every class, including those sold as “water-free”